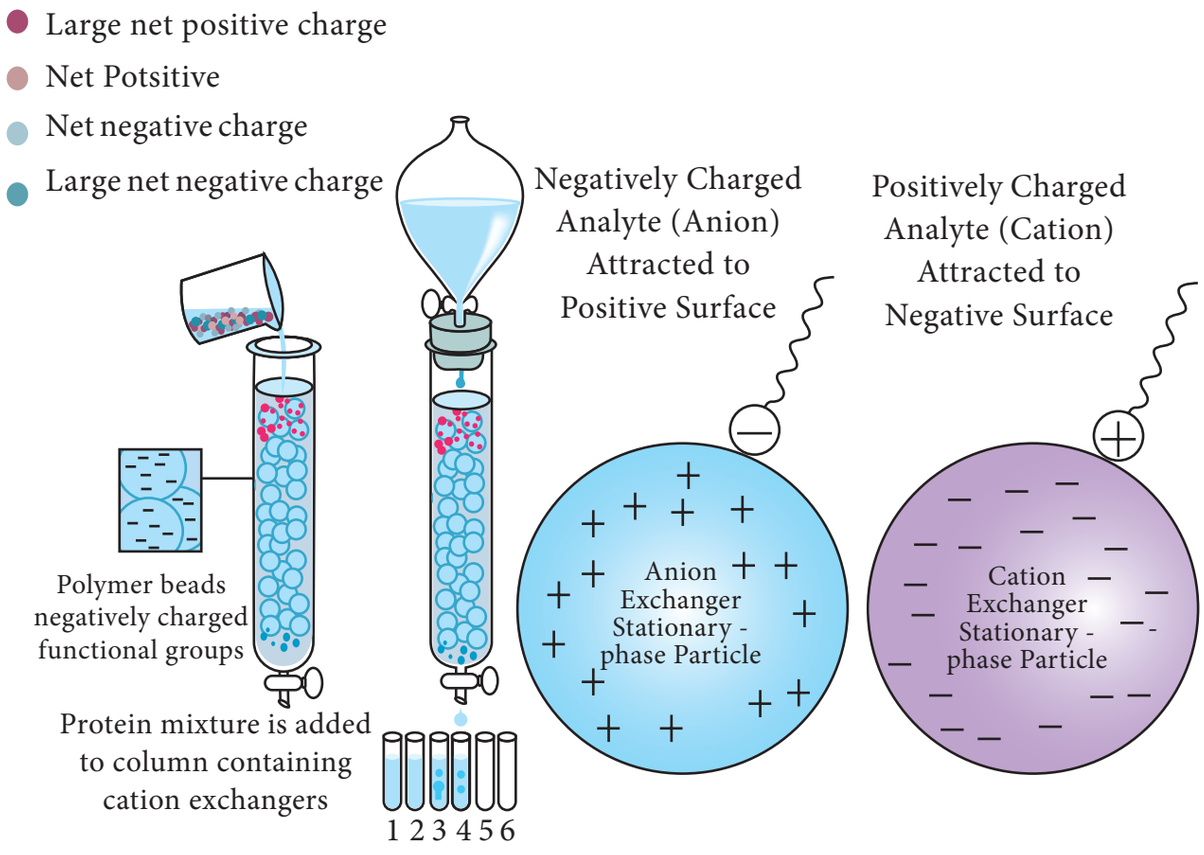
10.3.5 Ion-exchange chromatography

Principle : Ion exchange chromatography was developed by D'Alelio in 1942 using synthetic ion exchange medium based on polystyrene resin. Ion exchange chromatography is a displacement technique in which the ions in either the sample or the buffer displace the existing mobile ions associated with the fixed resin ions. Ion-exchange resins are insoluble, porous matrix containing numbers of ionic groups capable of binding ions of an opposite charge from the surrounding solution. As described in Fig.10.5, proteins move through the column depending upon the net charge at the pH being used.



Protein move through the column at rates determined by their net charge at the pH being used. With cation exchangers, proteins with a more negative net charge move faster and elute earlier.

Fig.10.5 Ion exchange chromatography

When the proteins are attached to the column material, increasing salt concentration or buffers with different pH are passed through the column. Proteins will be washed of the column at different times depending on their molecular structure, net charge and side group. Anion exchangers contain immobilized cationic groups that bind to anions and vice-versa. Examples of anion exchangers are Diethylaminoethyl (DEAE), aminoethyl (AE). Cationic exchangers include carboxymethyl (CM) cellulose, sulphopropyl (SP).

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